

THE UNIVERSITY OF MANCHESTER

Faculty of Biology, Medicine and Health

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# **Towards a Mechanistic Explanation of CHD–CKD**

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Do congenital heart disease and chronic kidney disease  
share a genetic basis?

**Research Project 1**

BIOL61230

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## DECLARATION

I declare that this report is my own work. No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

All data used in this project are publicly available and were obtained from the sources cited in Section 2. No individual-level human genotype data were accessed, and no ethical approval was required.

Where the work of others has been used, it is acknowledged and referenced. Analysis code written for this project is available at [github.com/sahoo2000/chd-ckd-genetics](https://github.com/sahoo2000/chd-ckd-genetics).

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Jagadrupa Sahoo

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Date

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## ACKNOWLEDGEMENTS

I would like to thank my supervisor, Dr David Talavera, for his guidance throughout this project, and in particular for pressing me to test whether the gene overlap I first observed was specific to the kidney rather than a general property of syndromic genes. That question changed the direction of the work and became its central result.

I am grateful to the teams behind the public resources this project depends on entirely: the Human Phenotype Ontology and Monarch Initiative, the HGNC, the Genome Aggregation Database, the Genebase team at the Broad Institute, the CKDGen Consortium, and FinnGen. It is worth recording that a project of this scope was possible on a laptop, without any controlled-access application, purely because these groups release their data openly.

Finally I thank the participants of UK Biobank, FinnGen and the CKDGen cohorts, whose contribution underlies every result reported here.

## ABSTRACT

Congenital heart disease (CHD) and chronic kidney disease (CKD) co-occur clinically, and this is widely assumed to reflect shared genetic causes. This study tested that assumption directly. Using curated gene–phenotype annotations from the Human Phenotype Ontology (332,599 annotations across 5,268 genes and 9,155 diseases), 1,660 genes causing a cardiac malformation and 1,198 causing a kidney malformation were found to overlap in 750 genes (odds ratio 5.81,  $p = 1.5 \times 10^{-144}$ ). However, this overlap proved largely non-specific: the kidney ranked only 16th of 194 organ systems, and CHD-associated diseases affect a median of 11 organ systems compared with 5 for other diseases. After adjusting for this pleiotropy in a disease-level logistic regression, the association with structural kidney malformation persisted (adjusted OR 1.77, 95% CI 1.52–2.07) whereas the association with chronic kidney disease vanished entirely (adjusted OR 0.95, 95% CI 0.73–1.23,  $p = 0.68$ ).

This dissociation was confirmed independently. Gene-burden statistics from 394,841 UK Biobank exomes showed that only dominant cystogenic genes (*PKD1*, *PKD2*, *IFT140*, *ALG9*) reach exome-wide significance for kidney traits, but that ciliary genes as a class carry a 9.7-fold excess of results at  $p < 0.001$  after those genes are removed, with 59 of 60 nominal results in the direction of worse kidney function ( $p = 5.3 \times 10^{-17}$ ). Signal tracked intraflagellar transport and dynein-2 rather than the BBSome. Cross-trait LD score regression using CKDGen ( $n \approx 981,000$ ) and FinnGen (4,270 CHD cases) found no genetic correlation between CHD and kidney function ( $r_g = -0.05$ ,  $p = 0.58$ ) against a positive control of  $r_g = -0.36$  ( $p = 2.0 \times 10^{-10}$ ) for eGFR and CKD.

Two further causal analyses were added. Two-sample Mendelian randomisation, using FinnGen CHD as exposure and CKDGen eGFR as outcome, found no causal effect (IVW  $-0.0009$ , 95% CI  $-0.0029$  to  $+0.0012$ ,  $p = 0.41$ ) but could only exclude effects larger than 0.0029 at 80% power, and a biologically impossible reverse direction returned null as required. Doubly robust estimation with machine-learning nuisance models showed the kidney-malformation association is robust to functional form ( $+3.8$  percentage points,  $p = 0.003$ ) but not to adjusting for organ-system identity, where a 13% positivity violation makes the comparison unreliable. The null result for CKD was stable across every specification.

The conclusion is that CHD and CKD do not share Mendelian genetic architecture once syndromic pleiotropy is accounted for. The genuine shared axis is between CHD and congenital anomalies of the kidney and urinary tract, mediated by ciliary biology.

**Keywords:** congenital heart disease, chronic kidney disease, ciliopathy, Human Phenotype Ontology, gene burden testing, LD score regression, pleiotropy

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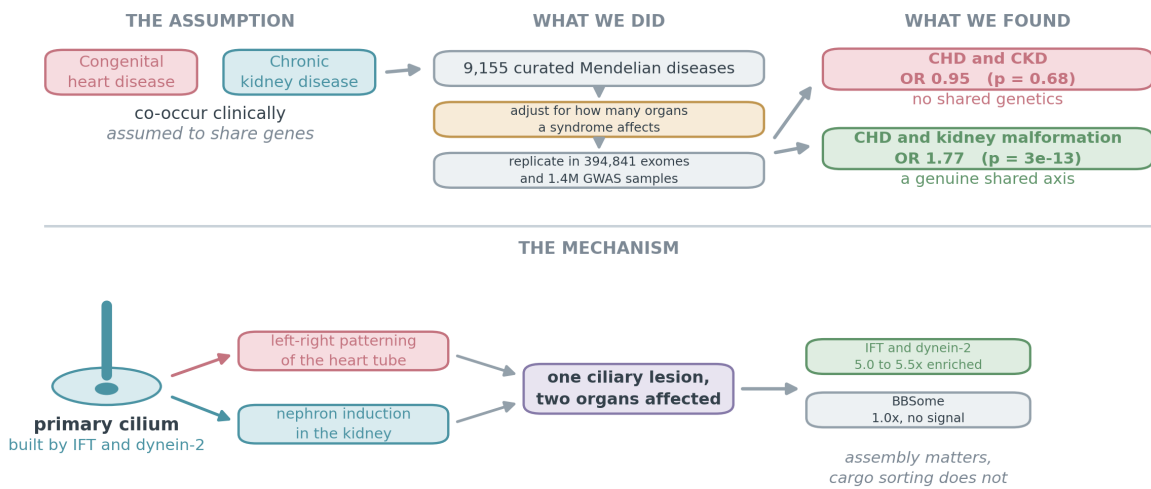
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## GRAPHICAL ABSTRACT

### Towards a Mechanistic Explanation of CHD-CKD



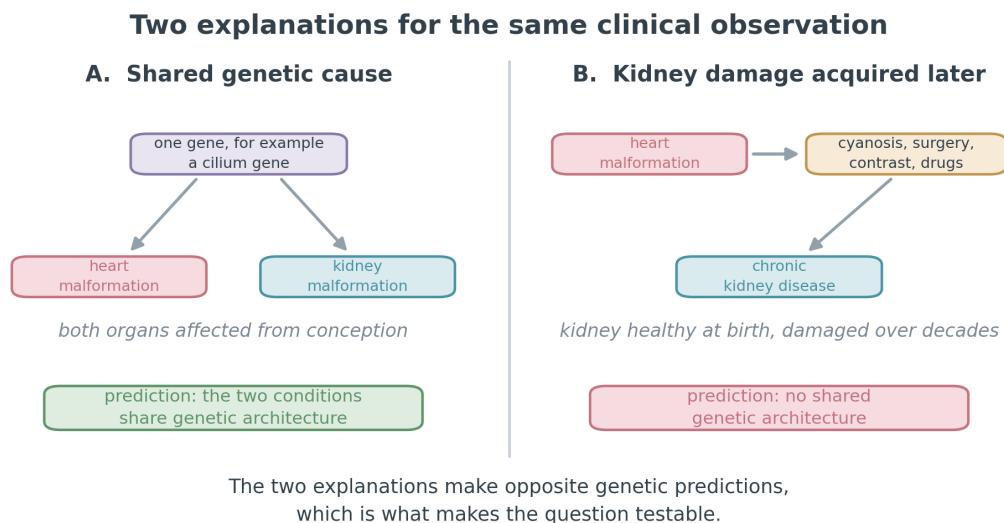
Rare loss-of-function burden in 394,841 UK Biobank exomes · 9.7x excess at  $p < 0.001$  · 59 of 60 in the predicted direction

# 1. INTRODUCTION

Congenital heart disease is the most common congenital anomaly, affecting approximately 1% of live births worldwide [1, 2]. It encompasses a spectrum of structural defects of the heart and great vessels, from small septal defects to complex malformations such as hypoplastic left heart syndrome [3]. Advances in cardiac surgery since the 1950s have transformed survival, and a growing population of adults now lives with congenital heart disease [4].

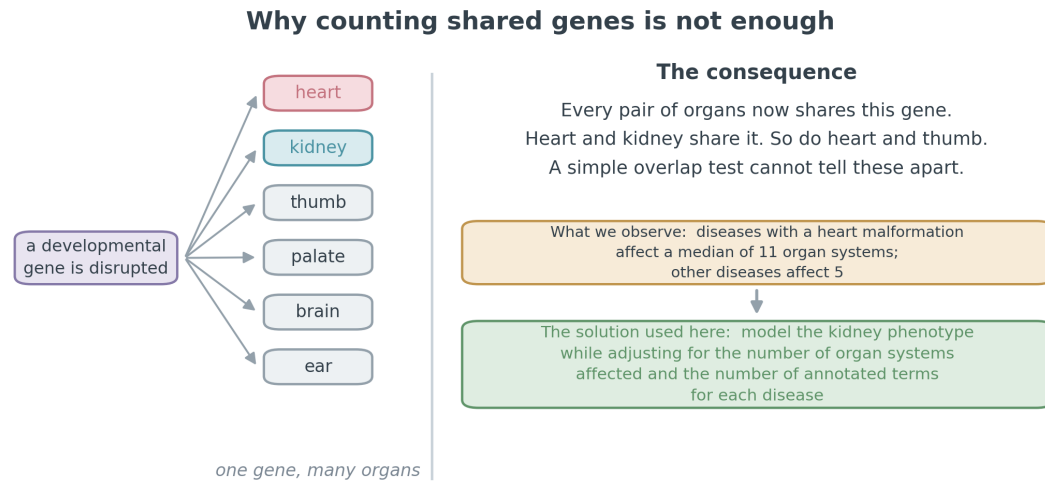
Chronic kidney disease is a recognised comorbidity in this population. Reported prevalence in adults with congenital heart disease is substantially higher than in the general population [5, 6], and kidney dysfunction predicts mortality in congenital heart disease cohorts [7]. Several mechanisms have been proposed: chronic cyanosis and altered haemodynamics, repeated exposure to iodinated contrast during catheterisation, cardiopulmonary bypass during surgical repair, and nephrotoxic medication [8, 9].

A distinct possibility is that the two conditions share genetic causes. This is biologically plausible. The primary cilium is required both for left–right patterning during cardiac looping and for normal nephron development, and ciliary gene defects produce syndromes affecting both organs [10, 11]. Several Mendelian syndromes—Alagille syndrome, the 22q11.2 deletion, Bardet–Biedl syndrome, nephronophthisis—feature cardiac and renal involvement together.



**Figure 1:** Two explanations for the same clinical observation. Under (A) a single gene disrupts both organs during development, so the two conditions share genetic architecture. Under (B) the kidney is normal at birth and is damaged over decades by the consequences of the cardiac lesion and its treatment. These make opposite genetic predictions, which is what makes the question testable.

However, plausibility is not evidence, and a specific methodological hazard applies. Genes causing congenital malformation syndromes are typically developmental regulators that disrupt many organs simultaneously. Any two organ systems will therefore appear to “share” genes simply because syndromic genes are pleiotropic. Distinguishing a genuine cardiorenal axis from this generic background requires an explicit control (Figure 2).



**Figure 2:** Why counting shared genes is not sufficient. A gene that disrupts early development affects many organs at once, so every pair of organs appears to share it. A simple overlap test cannot distinguish a genuine cardiorenal relationship from this background.

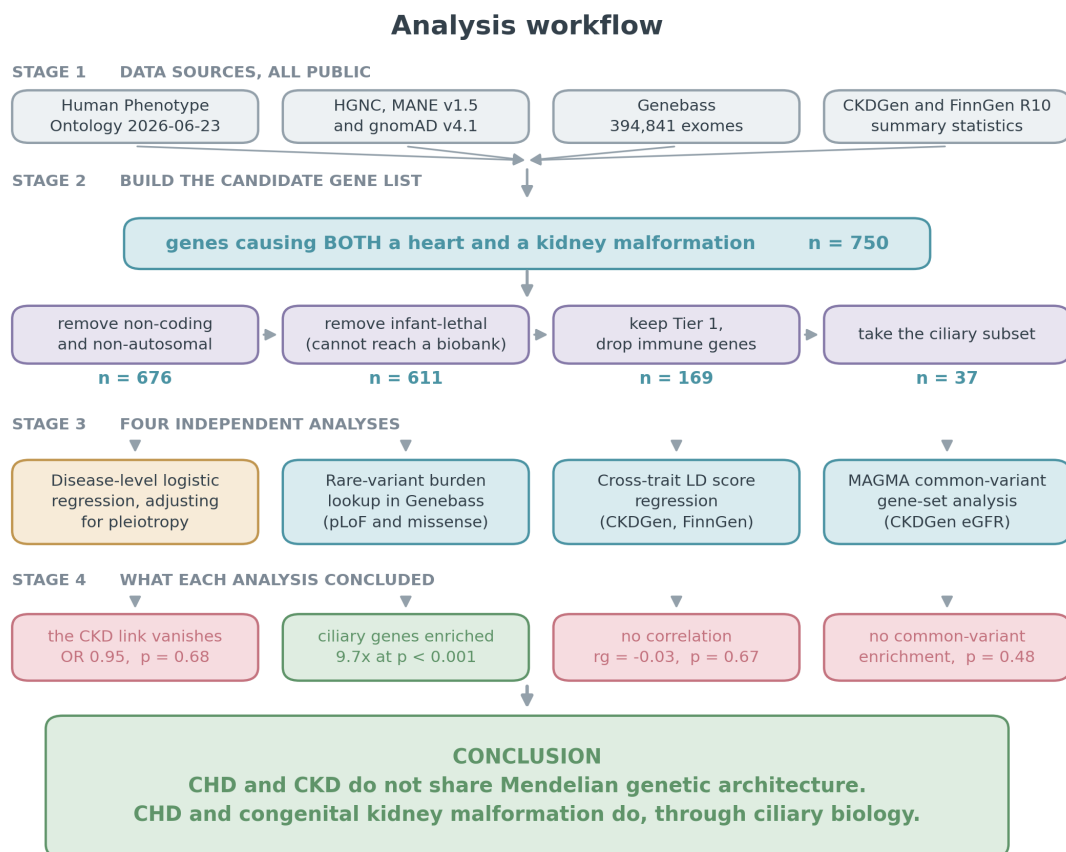
## 1.1 Aims

This study asked three questions:

1. Do congenital heart disease and kidney disease share genes more than expected by chance, and is any such overlap specific to the kidney?
2. Does any shared signal survive adjustment for syndromic pleiotropy, and does it concern structural kidney malformation or chronic kidney disease as a functional outcome?
3. Is the conclusion reproducible in independent data using a different variant class and a different statistical framework?

## 2. METHODOLOGY

All analyses used publicly available data. No individual-level genotype data were accessed, and no ethical approval was required. The overall workflow is shown in Figure 3.



**Figure 3:** Analysis workflow. Four independent analyses were applied to the same candidate gene list: a disease-level regression controlling for pleiotropy, a rare-variant burden lookup, cross-trait LD score regression, and a common-variant gene-set test.

## 2.1 Software versions and computing environment

All analyses were run on macOS (Darwin 25.5.0, Apple Silicon) using Python 3.9.6 with `pandas` 1.x, `numpy`, `scipy` and `statsmodels` 0.14.6. Figures were produced with `matplotlib` using the Agg backend at 300 dpi. Gene-level association from summary statistics used MAGMA v1.10 [26]. Table 1 lists every external resource with its version and access date.

**Table 1:** External data resources, versions and access routes. All are public; none required an application.

| Resource                 | Version / release | Content used                  | Access          |
|--------------------------|-------------------|-------------------------------|-----------------|
| Human Phenotype Ontology | 2026-06-23        | hp.obo, 20,413 terms          | Open (PURL)     |
| HPO annotations          | 2026-06-23        | 332,599 gene-phenotype rows   | Open (PURL)     |
| HGNC complete set        | current           | symbols, locus type, location | Open            |
| MANE Select              | v1.5              | GRCh38 gene coordinates       | NCBI FTP        |
| gnomAD constraint        | v4.1              | pLI, LOEUF (MANE transcripts) | Google Cloud    |
| Genebass                 | 394,841 exomes    | SKAT-O, burden, ACAT          | Open API        |
| CKDGen (Stanzick 2021)   | European          | eGFR, 8.84M variants          | Direct download |
| FinnGen                  | Release 10        | CHD, CKD, cystic kidney       | Direct download |
| 1000 Genomes             | Phase 3, EUR      | LD reference for MAGMA        | CTG Lab         |
| LD scores                | eur_w_ld_chr      | 1,290,028 HapMap 3 SNPs       | Zenodo          |
| MAGMA                    | v1.10             | gene and gene-set analysis    | CTG Lab         |

Analyses were re-run using the most recent releases of each resource available at the time of analysis rather than the versions current when the project was first conceived; version numbers are stated throughout so that any result can be reproduced exactly.

## 2.2 Curated gene-phenotype annotations

Gene-phenotype associations were obtained from the Human Phenotype Ontology (HPO), release 2026-06-23 [12]. The ontology file (hp.obo, 20,413 terms) and the gene-to-phenotype annotation file (332,599 annotations covering 5,268 genes and 9,155 diseases) were downloaded from the HPO project. These annotations derive from OMIM and Orphanet curation rather than automated text mining, which avoids the literature-popularity bias that affects text-mined resources.

The ontology was parsed into a directed acyclic graph and all descendant terms were resolved for three root terms:

- **Cardiac:** *Abnormal heart morphology* (HP:0001627), 333 descendant terms
- **Kidney (structural):** *Abnormal renal morphology* (HP:0012210), 472 terms
- **CKD (functional):** *Chronic kidney disease* (HP:0012622) and *Renal insufficiency* (HP:0000083), 15 terms combined

## 2.3 Testing overlap and specificity

Gene-set overlap was tested with a one-sided Fisher exact test against a background of all HPO-annotated disease genes. To assess whether any overlap was specific to the kidney, the same test was repeated against all 194 non-cardiovascular organ-morphology term sets in the ontology

containing at least 150 genes. Terms descending from *Abnormality of the cardiovascular system* (HP:0001626) were excluded to avoid trivial self-overlap.

## 2.4 Adjusting for syndromic pleiotropy

Analysis was moved to the disease level. For each of the 9,155 diseases, binary indicators were derived for cardiac malformation, structural kidney anomaly and CKD, together with two measures of pleiotropy: the number of the 23 top-level HPO organ systems affected, and the total number of annotated terms. Logistic regression was then fitted:

$$\text{logit } P(\text{kidney phenotype}) = \beta_0 + \beta_1 \cdot \text{CHD} + \beta_2 \cdot n_{\text{systems}} + \beta_3 \cdot n_{\text{terms}} \quad (1)$$

where  $\exp(\beta_1)$  is the adjusted odds ratio of interest. The same model was fitted separately for each of twelve specific cardiac lesions.

## 2.5 Causal interpretation and its assumptions

The adjustment above is not merely a statistical convenience; it is an attempt at causal identification, and it is worth stating what has to be true for the adjusted odds ratio to mean anything causal. Figure 4 gives the assumed causal structure.

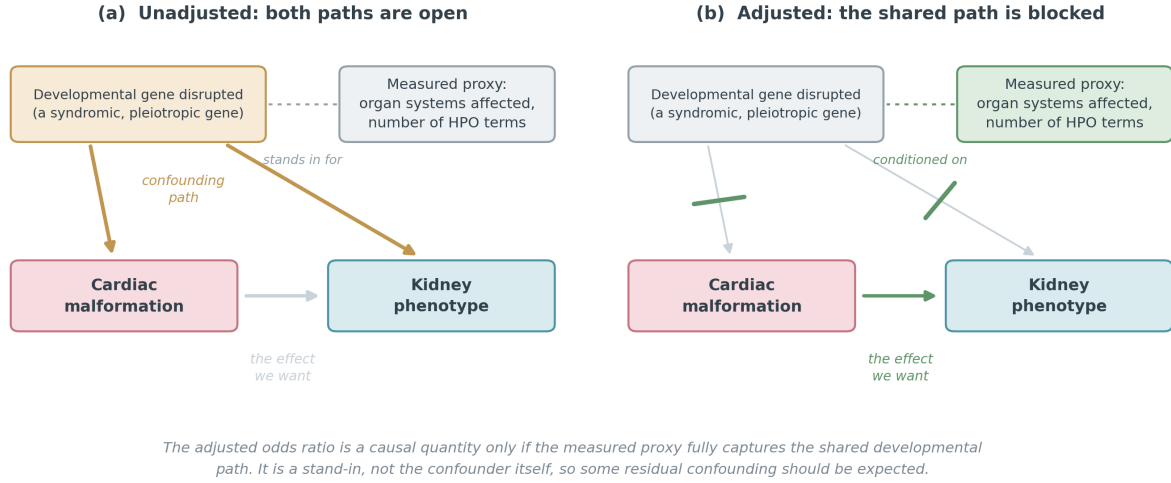
The quantity of interest is the effect of cardiac malformation on kidney phenotype. A confounding (backdoor) path runs from a disrupted developmental gene to both organs, because syndromic genes are pleiotropic by definition. If that path is left open, the two organs appear associated even when neither causes the other, and the unadjusted odds ratio of 5.81 is largely a measurement of that path rather than of any cardiorenal axis.

Conditioning on the pleiotropy measures is intended to block it. Three assumptions are required, and only the first is fully satisfied here:

1. **No unmeasured confounding given the proxy.** Organ-system count and term count must capture the shared developmental path completely. They are summaries of a latent quantity, not the quantity itself, so this assumption is certainly violated to some degree and residual confounding should be expected.
2. **Correct functional form.** Logistic regression forces the effect of organ-system count to be linear on the log-odds scale. This is tested directly in Section 2.11.
3. **Positivity.** Every disease must have a non-negligible chance of both affecting and not affecting the heart, given its covariates. This is also assessed in Section 2.11, where it turns out to be the binding constraint.

Two variables are deliberately excluded from every adjustment set. The cardiovascular organ system is what the exposure is, so conditioning on it would remove the exposure itself. The genitourinary system is where the outcome lives, so conditioning on it would block the very

effect being estimated. Including either would force a null result for reasons that have nothing to do with biology.



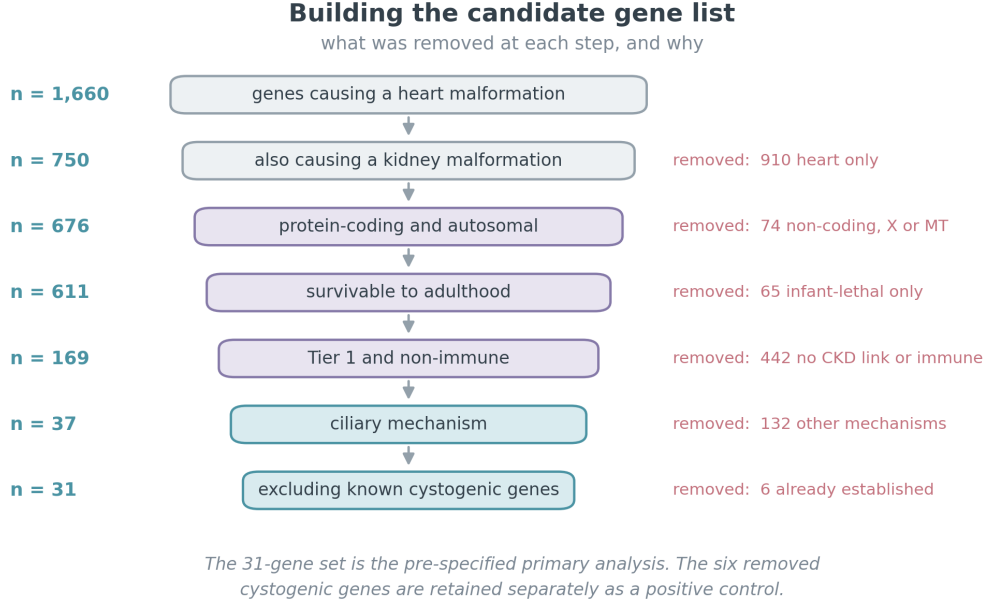
**Figure 4:** The causal model behind the pleiotropy adjustment. (a) Unadjusted, the confounding path through the shared developmental gene is open and inflates the apparent association. (b) Conditioning on the measured proxy is intended to block it. The proxy is a stand-in for the confounder rather than the confounder itself, which is the main threat to a causal reading of the adjusted estimate.

## 2.6 Candidate gene list construction

Genes annotated to both a cardiac and a kidney malformation term were retained (750 genes), then annotated with current approved symbols, chromosomal location and locus type from HGNC [13], genomic coordinates from MANE Select v1.5 [14], and loss-of-function constraint (pLI and LOEUF) from gnomAD v4.1 [15]. Non-coding and non-autosomal genes were removed.

A survivability filter was then applied. Genes whose only documented phenotypes included perinatal or infant lethality (HP:0001522, HP:0003811, HP:0003826, HP:0034241, HP:0003819) with no adult- or juvenile-onset disease on record were excluded, because affected individuals cannot appear in an adult cohort. This removed 65 genes, leaving 611.

Genes were then tiered by whether they additionally had documented CKD or renal insufficiency. Tier 1 (178 genes) required all three. Immune-mediated genes entering through autoimmune disease (HLA loci, *PTPN22*, *STAT4*, *MEFV*, *IL6*, *IL10*, *TLR4*, *HFE*, *SAA1*, *TNXB*) were removed as mechanistically distinct, giving 169 genes. A 37-gene ciliary subset was defined from the ciliopathy, nephronophthisis and polycystic kidney disease mechanism classes. Six of these are already established causes of cystic kidney disease (*PKD1*, *PKD2*, *IFT140*, *ALG9*, *GANAB*, *DNAJB11*); because including them in a pooled test would guarantee a positive result that merely rediscovers autosomal dominant polycystic kidney disease, they were held out as a positive control and the remaining 31 genes pre-specified as the primary set. Figure 5 summarises the filtering.



**Figure 5:** Construction of the candidate gene list, showing how many genes were removed at each step and why. Bar widths are proportional to  $\log_{10}(n)$ .

## 2.7 Gene burden statistics from exome sequencing

Rare-variant gene-burden association statistics were obtained from GeneBass [16], which provides SKAT-O [17], burden and ACAT [18] tests across 4,529 phenotypes in 394,841 UK Biobank exomes [19, 20]. Results were retrieved programmatically for each candidate gene and filtered to twenty renal phenotypes: five continuous biomarkers (creatinine, cystatin C, urea, microalbumin, urinary creatinine) and fifteen ICD-10 based endpoints. Both the predicted loss-of-function (pLoF) and missense burden sets were retrieved, the latter serving as a negative control.

Enrichment was assessed by comparing observed counts of nominally significant results against the number expected under the null. Because individual genes were underpowered, a directional test was also applied: among results with  $p < 0.05$ , the proportion with a positive effect estimate (worse kidney function) was compared against 0.5 with an exact binomial test. Under the null this proportion is 0.5 regardless of selection on  $p$ .

## 2.8 Cross-trait LD score regression

Genetic correlation was estimated by cross-trait LD score regression [21, 22]. Summary statistics were obtained for estimated glomerular filtration rate from CKDGen (European-ancestry meta-analysis, 8.8 million variants, mean  $n = 980,825$ ) [23], and for congenital malformations of the heart and great arteries (4,270 cases), chronic kidney disease (10,039 cases) and cystic kidney disease (947 cases) from FinnGen release 10 [24].

Variants were restricted to the HapMap 3 subset used for the pre-computed European LD scores (1,290,028 SNPs,  $M_{5,50} = 1,173,569$ ) derived from 1000 Genomes Phase 3 [25]. Strand-



ambiguous variants, those with minor allele frequency below 0.01, and those with  $\chi^2 > \max(80, 0.001N)$  were removed. Effective sample size  $N_{\text{eff}} = 4/(1/n_{\text{case}} + 1/n_{\text{control}})$  was used for binary traits.

Heritability was estimated from the regression  $\mathbb{E}[\chi_j^2] = 1 + (Nh^2/M)\ell_j$ , and genetic covariance from  $\mathbb{E}[z_{1j}z_{2j}] = (\sqrt{N_1N_2}\rho_g/M)\ell_j + a$ , where the intercept  $a$  absorbs sample overlap. Standard errors came from a 200-block genome-ordered jackknife. Because the reference implementation requires Python 2, which no longer installs cleanly, the estimator was reimplemented; it was validated against a published heritability estimate and against a positive control trait pair before the results below were interpreted.

## 2.9 Common-variant gene and gene-set analysis

Gene-level association was computed with MAGMA v1.10 [26] from the CKDGen European-ancestry eGFR meta-analysis. SNP identifiers, chromosomes and build-37 positions were extracted for all 8,842,706 variants carrying an rs identifier. SNPs were assigned to genes using the NCBI37.3 gene location file with a zero-kilobase window, so only variants within the transcribed region contribute; 18,038 of 19,427 genes received at least one SNP. Gene  $p$ -values used the default SNP-wise mean model with the 1000 Genomes Phase 3 European panel (503 individuals) as LD reference.

Gene-set analysis used MAGMA’s competitive test, which asks whether genes in a set have smaller  $p$ -values than genes outside it, conditioning internally on gene size, gene density, sample size and minor allele count together with the logarithms of each. Four sets were tested: all 37 ciliary genes, the 31 with cystogenic genes removed, the 6 cystogenic genes alone, and the 166 Tier 1 genes with a gene-location entry. Gene symbols were mapped to Entrez identifiers through HGNC; all four sets mapped completely.

## 2.10 Mendelian randomisation

Genetic variants are allocated at conception and precede all postnatal exposures, so variants that raise liability to congenital heart disease provide a natural experiment for asking whether that liability causes later kidney impairment. Two-sample Mendelian randomisation was performed with FinnGen R10 CONGEN\_HEART\_ARTER (4,270 cases, 407,911 controls) as exposure and the CKDGen eGFR meta-analysis as outcome.

The FinnGen chronic kidney disease GWAS was deliberately *not* used as the outcome despite being available. Exposure and outcome would then be drawn from the same individuals, and under complete sample overlap weak-instrument bias pulls the estimate towards the confounded observational association — precisely the artefact this report argues against. CKDGen is a largely non-Finnish meta-analysis, so overlap is minimal.

Instruments were selected at two thresholds:  $p < 5 \times 10^{-8}$ , and a relaxed  $p < 10^{-5}$  to increase instrument count at the cost of instrument strength. Because CKDGen is reported on GRCh37 and FinnGen R10 on GRCh38, the two studies were merged on rsID rather than position. Independent signals were retained by distance-based clumping, keeping the strongest variant

within each 500 kb window; a linkage-disequilibrium reference panel was not used, so residual correlation between instruments cannot be excluded. Variants with minor allele frequency below 1% were discarded, which removed four ultra-rare Finnish-enriched 17q25 signals whose odds ratios of 3.5–4.2 rest on fewer than ten carrier alleles.

Alleles were harmonised to the exposure effect allele, with strand flips resolved by complementation and palindromic variants resolved by allele frequency, or dropped when the frequency fell between 0.42 and 0.58. Causal estimates used inverse-variance weighting with multiplicative random effects, the weighted median (standard error from 1,000 bootstrap replicates) and MR-Egger, whose intercept was used as a test for directional pleiotropy. Heterogeneity was quantified by Cochran’s  $Q$ , influence by leave-one-out, and instrument strength by the mean  $F$ -statistic and the summed variance explained on the liability scale.

Because a null result is only interpretable alongside a statement of power, the minimum detectable effect at 80% power was computed as  $2.80 \times \text{SE}$ .

The reverse direction was run as a negative control. A cardiac malformation is complete by approximately the eighth week of gestation, decades before any adult measurement of kidney function, so adult eGFR cannot cause congenital heart disease. The true answer is known to be null, and any departure from null would indicate a fault in the pipeline or in the instruments rather than a biological finding. Directionality was additionally assessed with the Steiger test. Because summing per-variant  $r^2$  across hundreds of instruments accumulates noise in the outcome trait, the verdict was taken from the number of instruments individually favouring each direction rather than from the summed values.

## 2.11 Doubly robust sensitivity analysis

Two of the three assumptions listed in Section 2.5 — functional form and positivity — were tested using augmented inverse probability weighting (AIPW) with machine learning nuisance models.

The machine learning here is not used for prediction, and no predictive accuracy is reported. Gradient boosting is used only as a flexible description of the confounders, inside an estimator that still targets the same causal contrast. Two models are fitted: a propensity model for  $P(\text{cardiac} \mid X)$  and an outcome model for  $P(\text{kidney} \mid \text{cardiac}, X)$ . AIPW combines them so that the estimate remains consistent if either one is correctly specified.

Both nuisance models were histogram gradient-boosting classifiers (200 iterations, learning rate 0.1, maximum depth 4) fitted with five-fold cross-fitting stratified jointly on exposure and outcome, so that no disease contributed to fitting the model that later made predictions about it. Propensity scores were truncated to  $[0.01, 0.99]$  and the truncated fraction is reported as a positivity diagnostic. The estimand is a risk difference, not an odds ratio, so it is not numerically comparable to  $\exp(\beta_1)$ .

Two adjustment sets were used. The first contains only organ-system count and term count, so that the sole difference from the primary analysis is the shape of the model. The second adds

one indicator per organ system, so that the adjustment responds to *which* systems are involved rather than only how many. Cardiovascular and genitourinary indicators were excluded from both, for the reasons given above, leaving 21 organ-system columns.

### 2.12 Multiple testing

Three correction strategies were used, chosen to match each question.

For the HPO organ-specificity screen (194 tests) and the disease-level regressions, results are reported with exact confidence intervals and  $p$ -values without correction, because the comparison of interest is between two pre-specified models rather than a search across many.

For the gene-burden lookups, two thresholds are quoted: Bonferroni across the tests performed ( $0.05/706 = 7.1 \times 10^{-5}$  for the ciliary set,  $0.05/3226 = 1.5 \times 10^{-5}$  for Tier 1), and the conventional exome-wide threshold for gene-based tests of  $2.5 \times 10^{-6}$ . Benjamini–Hochberg  $q$ -values [27] are additionally reported. The Benjamini–Hochberg step-up procedure is implemented as  $q_{(i)} = \min_{j \geq i} \left( \frac{n}{j} p_{(j)} \right)$ , evaluated by a cumulative minimum running from the largest  $p$ -value downwards; the implementation was checked against `statsmodels.stats.multitest`.

For the MAGMA gene-set analysis, four sets were tested and no result survives Bonferroni correction at  $0.05/4 = 0.0125$ .

### 2.13 Directional testing

Because individual genes in the candidate set are underpowered, a test with greater sensitivity was applied. Among gene  $\times$  phenotype results with  $p < 0.05$ , the proportion with a positive burden effect (indicating worse kidney function) was compared against 0.5 with an exact binomial test. This is valid despite selection on  $p$ : under the null hypothesis the sign of an effect estimate is independent of its magnitude, so selected results remain equally likely to point either way. An empirical null was also constructed by drawing 200,000 samples from  $\text{Binomial}(n, 0.5)$  to confirm the analytic  $p$ -value.

### 2.14 Reproducibility

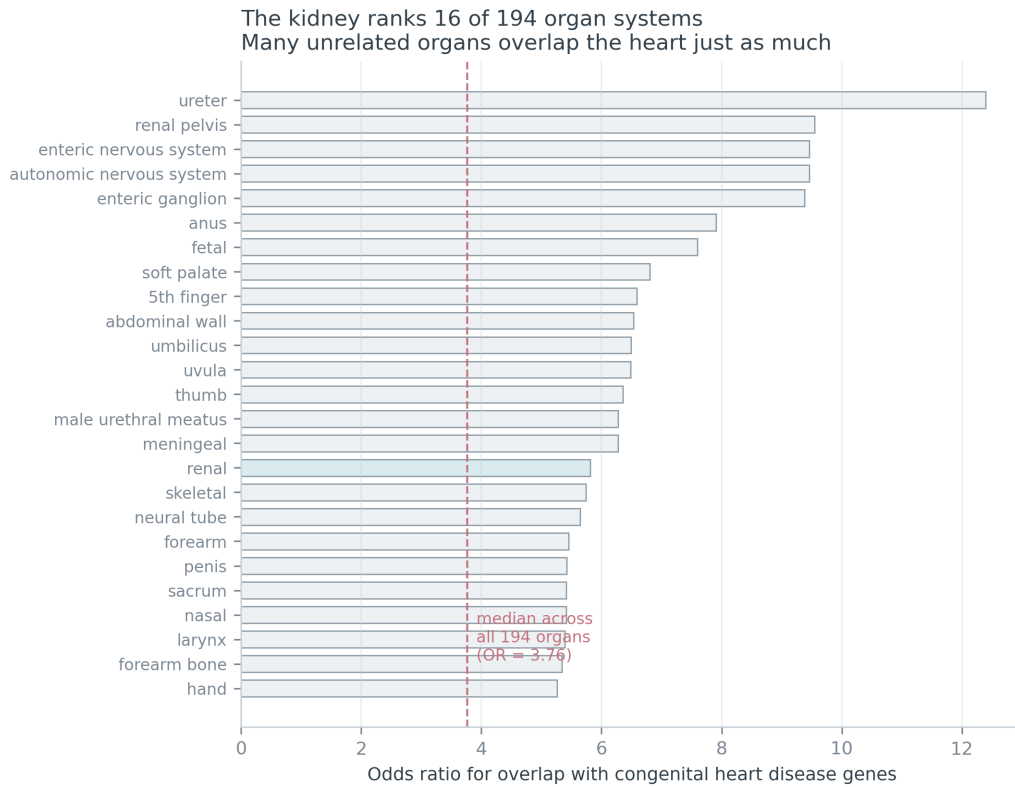
All analysis code is available at [github.com/sahoo2000/chd-ckd-genetics](https://github.com/sahoo2000/chd-ckd-genetics), organised as numbered scripts to be run in order. Every result table and figure in this report is regenerated by that code from the public sources in Table 1. Intermediate result files are included in the repository so that figures can be redrawn without repeating the slow steps; the MAGMA gene analysis in particular takes approximately four hours.

### 3. RESULTS

#### 3.1 Cardiac and kidney genes overlap, but not specifically

Of the genes with curated disease annotations, 1,660 cause a cardiac malformation and 1,198 cause a kidney malformation. The 750-gene overlap is far beyond chance (OR 5.81,  $p = 1.5 \times 10^{-144}$ ).

This overlap is not specific to the kidney. Repeating the identical test against 194 non-cardiovascular organ systems placed the kidney only 16th (Figure 6). The median odds ratio across all organ systems was 3.76, so the kidney sits only 1.5 times above a typical organ. Thumb (6.36), uvula (6.49), soft palate (6.80) and abdominal wall (6.53) morphology all overlap cardiac genes more strongly than the kidney does.



**Figure 6:** Overlap of congenital heart disease genes with each organ system. The kidney (highlighted) ranks 16th of 194. Many anatomically unrelated structures overlap the heart as strongly, which is the signature of syndromic pleiotropy rather than a specific cardiorenal axis.

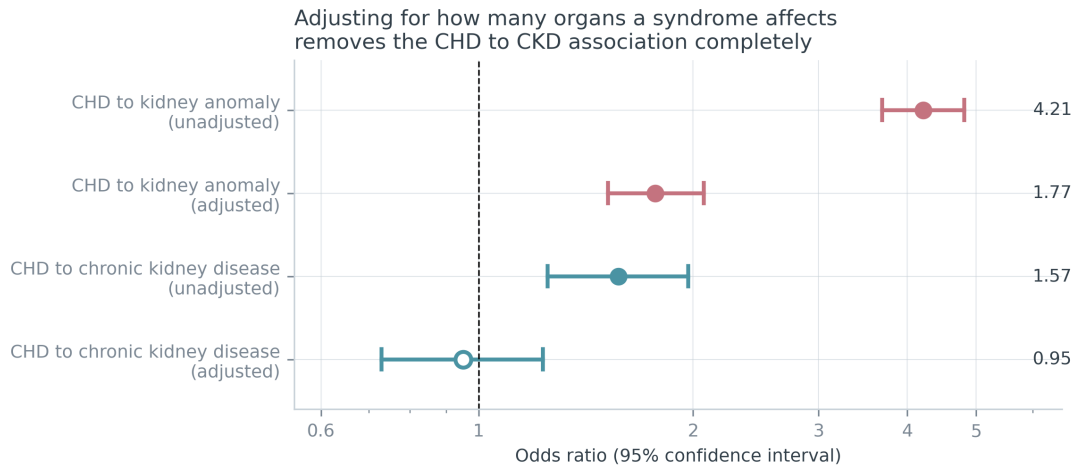
The explanation is pleiotropy. Diseases involving a cardiac malformation affect a median of 11 organ systems, against 5 for all other diseases (Mann–Whitney  $p = 5.3 \times 10^{-297}$ ). Genes disrupting early development disrupt many organs at once.

### 3.2 Adjustment separates two different questions

After adjusting for the number of organ systems affected and the number of annotated terms, the two putative associations behave completely differently (Figure 7; Table 2).

**Table 2:** Disease-level association between congenital heart disease and kidney phenotypes, before and after adjustment for pleiotropy ( $n = 9,155$  diseases).

| Outcome                   | Model      | OR   | 95% CI    | $p$                   |
|---------------------------|------------|------|-----------|-----------------------|
| Structural kidney anomaly | Unadjusted | 4.21 | 3.69–4.81 | $2.0 \times 10^{-94}$ |
| Structural kidney anomaly | Adjusted   | 1.77 | 1.52–2.07 | $3.1 \times 10^{-13}$ |
| Chronic kidney disease    | Unadjusted | 1.57 | 1.25–1.97 | $1.3 \times 10^{-4}$  |
| Chronic kidney disease    | Adjusted   | 0.95 | 0.73–1.23 | 0.68                  |

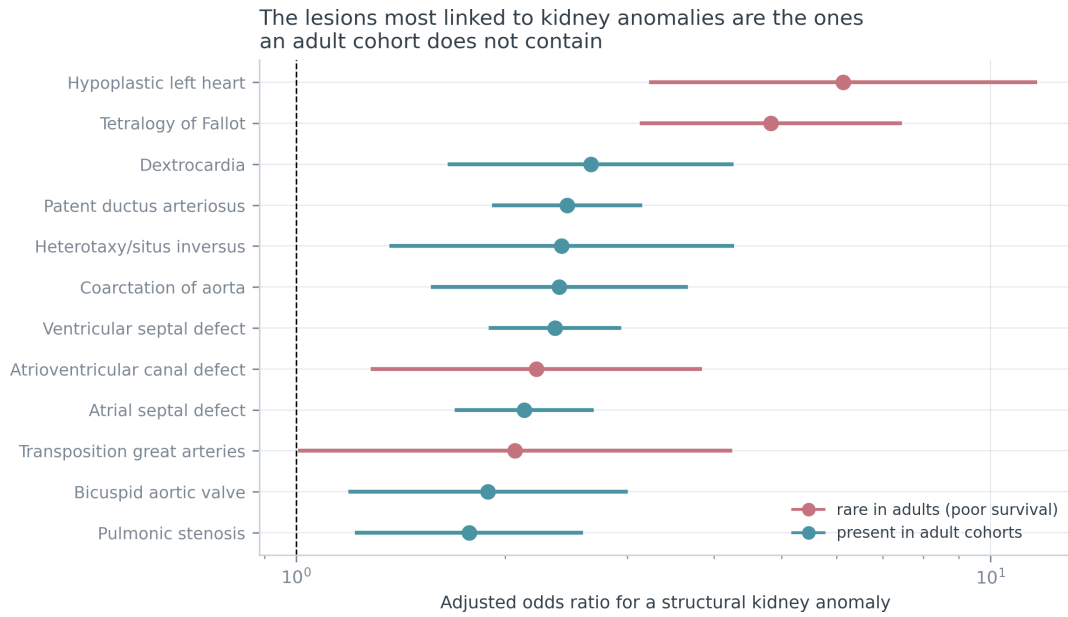


**Figure 7:** Odds ratios with 95% confidence intervals. The association with structural kidney malformation survives adjustment; the association with chronic kidney disease does not, and its adjusted interval spans the null.

The association with structural kidney malformation is attenuated but robust. The association with chronic kidney disease disappears entirely: the naive odds ratio of 1.57 is fully explained by the fact that heart-malformation syndromes affect many organs.

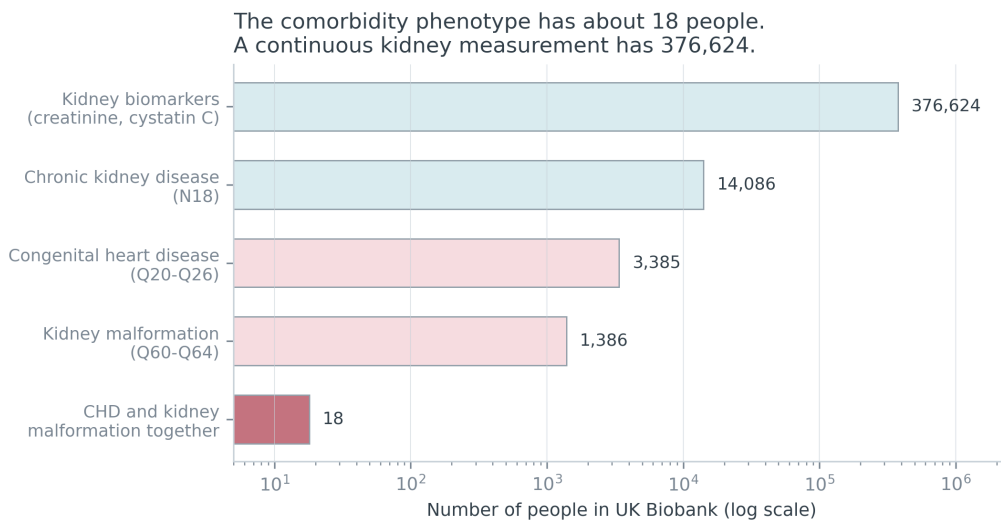
### 3.3 The lesions carrying the signal are absent from adult cohorts

Testing each cardiac lesion separately (Figure 8) showed that the strongest associations with kidney anomaly come from hypoplastic left heart syndrome (adjusted OR 6.13, 95% CI 3.24–11.6), tetralogy of Fallot (4.83, 3.15–7.41) and atrioventricular canal defect (2.22, 1.29–3.82). These are precisely the lesions that an adult cohort recruited at ages 40–69 will barely contain. The lesions that such a cohort does contain in numbers—bicuspid aortic valve (1.89), pulmonic stenosis (1.77), atrial septal defect (2.13)—sit at the bottom of the ranking.



**Figure 8:** Adjusted odds ratio for a structural kidney anomaly, by cardiac lesion. The ordering runs almost exactly opposite to availability in an adult biobank.

This has a direct consequence for study design. In UK Biobank there are approximately 3,385 congenital heart disease cases and 1,386 cases of congenital kidney anomaly among roughly 450,000 participants. If independent, about 10 individuals would carry both; applying the observed enrichment gives roughly 18 (Figure 9). No burden test has power on 18 cases at any effect size. A continuous kidney measurement, by contrast, is available in 376,624 participants.



**Figure 9:** Available sample sizes in UK Biobank. The comorbidity phenotype is not analysable; a quantitative kidney trait is measured in essentially everyone.

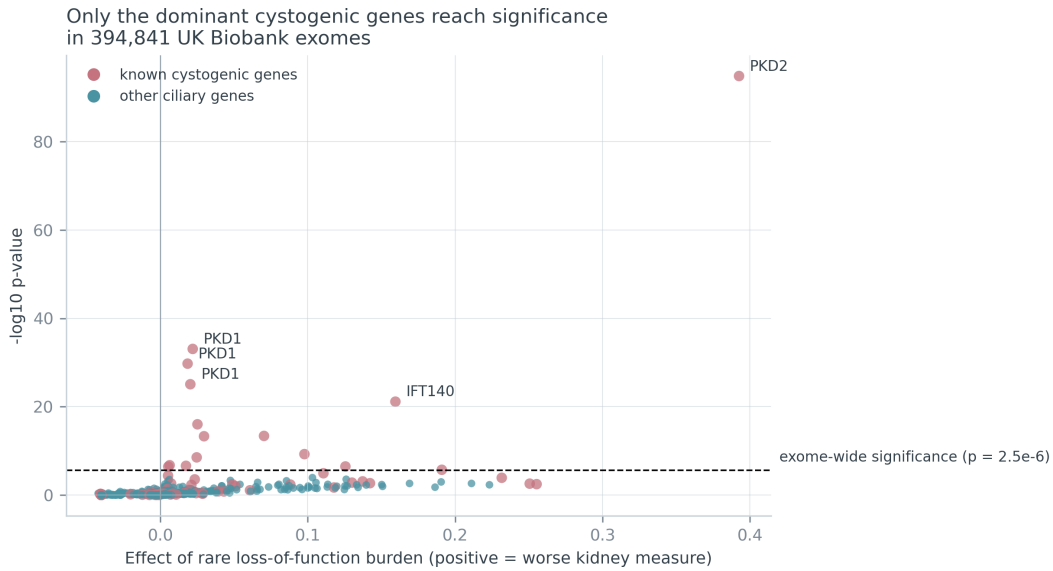
### 3.4 Independent replication in 394,841 exomes

Gene-burden statistics were retrieved for 165 Tier 1 genes, giving 3,226 gene  $\times$  renal phenotype associations. Four genes reach exome-wide significance ( $p < 2.5 \times 10^{-6}$ ), and all four are

dominant, adult-onset cystogenic genes (Table 3; Figure 10). The 31 recessive ciliopathy genes—*NPHP1*, *NPHP3*, *CEP290*, the *BBS* family, *TMEM67*, *MKS1*—reach nothing, exactly as the survivability argument predicts.

**Table 3:** Exome-wide significant gene  $\times$  renal phenotype associations, pLoF burden, SKAT-O.

| Gene          | Phenotype                  | $p$                   | $\beta$ |
|---------------|----------------------------|-----------------------|---------|
| <i>PKD2</i>   | Q61 cystic kidney disease  | $1.3 \times 10^{-95}$ | 0.393   |
| <i>PKD1</i>   | Cystatin C                 | $8.6 \times 10^{-34}$ | 0.022   |
| <i>PKD1</i>   | Creatinine                 | $1.9 \times 10^{-30}$ | 0.019   |
| <i>PKD1</i>   | Urea                       | $8.5 \times 10^{-26}$ | 0.020   |
| <i>IFT140</i> | Q61 cystic kidney disease  | $7.4 \times 10^{-22}$ | 0.160   |
| <i>PKD2</i>   | Creatinine                 | $1.0 \times 10^{-16}$ | 0.025   |
| <i>IFT140</i> | N28 other kidney disorders | $4.3 \times 10^{-14}$ | 0.070   |
| <i>PKD2</i>   | N18 chronic renal failure  | $5.7 \times 10^{-10}$ | 0.098   |
| <i>ALG9</i>   | Q61 cystic kidney disease  | $2.0 \times 10^{-6}$  | 0.191   |



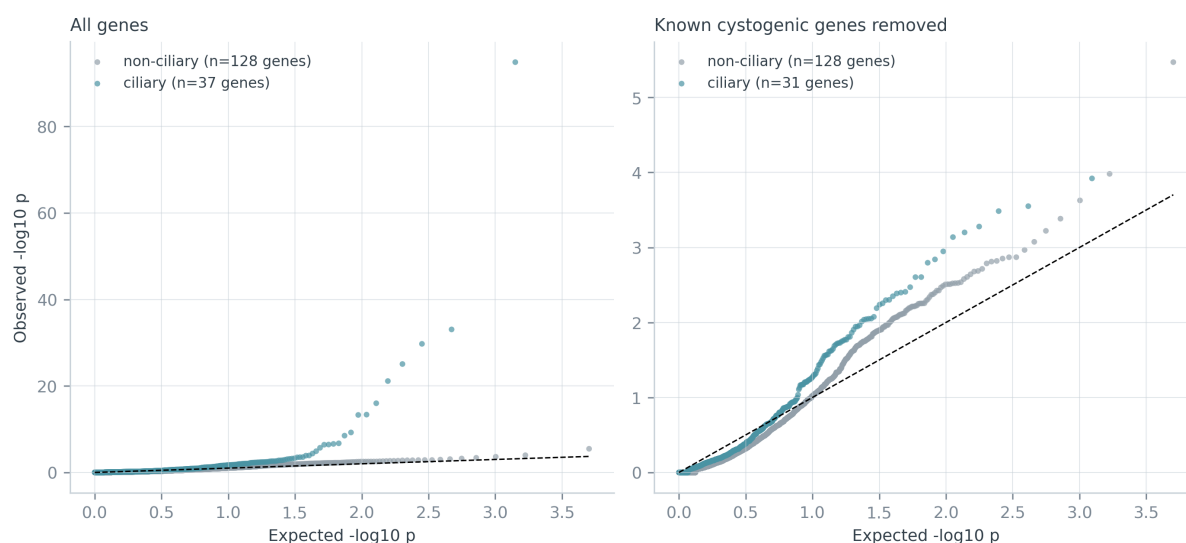
**Figure 10:** Gene-burden associations for the 37 ciliary genes across 20 renal phenotypes. Only dominant cystogenic genes clear exome-wide significance.

Critically, signal remains after those genes are removed. Across the remaining 620 tests there were 27 results below  $p = 0.01$  against 6.2 expected, and 6 below  $p = 0.001$  against 0.6—a 9.7-fold excess. The direction is near-unanimous: 59 of 60 nominally significant residual results show worse kidney function (sign test  $p = 5.3 \times 10^{-17}$ ). Across all 706 tests direction runs at 44% positive, confirming that this is signal concentrated in the tail rather than a systematic artefact.

Widening the analysis from 37 ciliary genes to all 165 Tier 1 genes added no further significant genes and diluted the excess (Figure 11). A ciliary gene is 3.2 times more likely to carry a  $p < 0.01$  renal association than a non-ciliary Tier 1 gene (Fisher OR 3.20,  $p = 1.6 \times 10^{-9}$ ). On

congenital kidney malformation (Q63), 11 of 35 ciliary genes are nominally significant (31%) against 6 of 126 non-ciliary genes (5%, exactly chance).

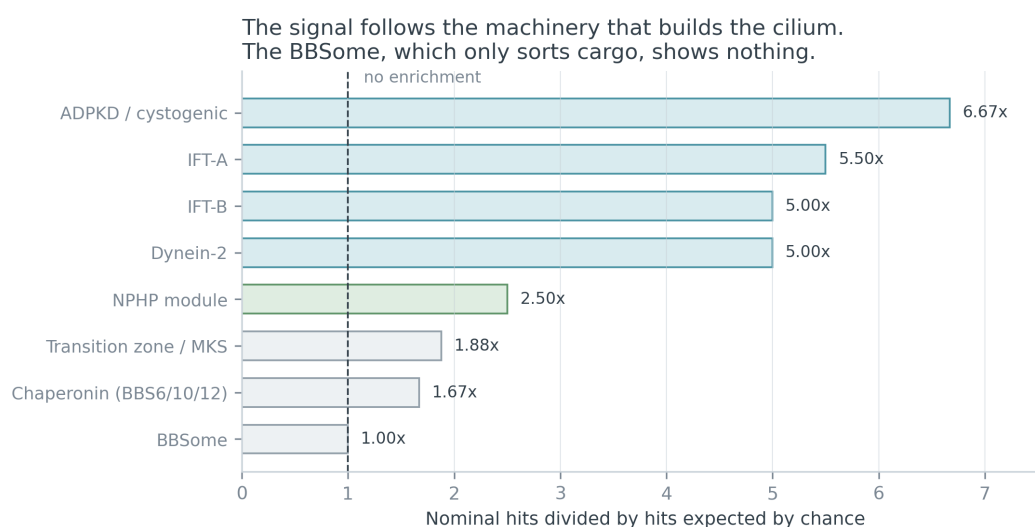
Ciliary genes depart from the null even after the known cystogenic genes are removed



**Figure 11:** Quantile–quantile plots of gene-burden  $p$ -values. Ciliary genes depart from the null distribution more than non-ciliary candidate genes, and continue to do so after the known cystogenic genes are removed (right panel).

### 3.5 Signal follows cilium assembly, not cargo sorting

Grouping genes by ciliary subcomplex revealed a coherent gradient (Figure 12). Intraflagellar transport complexes A and B and dynein-2—the machinery that assembles and maintains the cilium—show 4- to 5.5-fold enrichment. The BBSome, which sorts cargo through an already-assembled cilium, sits at exactly the null expectation. This is not explained by statistical power: within the ciliary set, genes with more nominal hits do not have more qualifying variants (Mann–Whitney  $p = 0.078$ ).



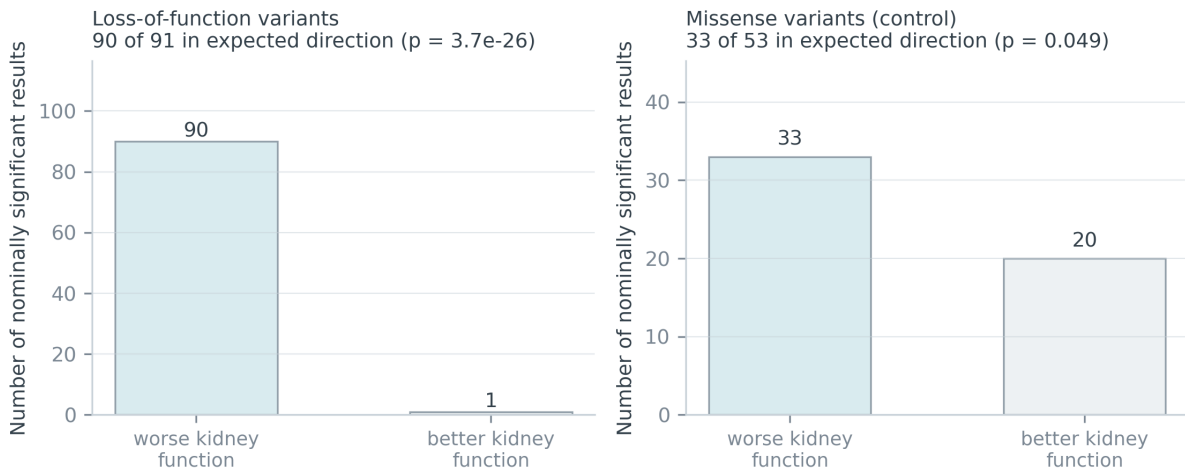
**Figure 12:** Enrichment of nominally significant renal associations by ciliary subcomplex.



### 3.6 A negative control

Repeating the analysis on the unfiltered missense burden set over the same genes and phenotypes reproduced none of the above (Figure 13). The excess at  $p < 0.001$  fell from 9.7-fold to 1.6-fold, and directional consistency fell from 98% to 67%. The single surviving association, *PKD1* with creatinine ( $p = 2.6 \times 10^{-14}$ ), has a burden effect of  $-0.0005$ —indistinguishable from zero and pointing the wrong way, indicating a variance-driven SKAT-O result rather than a consistent shift. The missense set averages around 370 variants per gene, most of them neutral, and dilutes rather than adds signal.

Loss-of-function results point one way; unfiltered missense results do not



**Figure 13:** Direction of effect among nominally significant results, with all 37 ciliary genes included. Loss-of-function results point overwhelmingly toward worse kidney function (90 of 91); unfiltered missense results do not (33 of 53). The corresponding figures with the six known cystogenic genes removed are 59 of 60 and 28 of 42 respectively, as reported in the text.

### 3.7 Common-variant evidence points elsewhere

Gene-level association from common variants was computed with MAGMA using the CKDGen eGFR meta-analysis and a 1000 Genomes European reference panel, annotating 8.84 million variants to 18,038 genes. A competitive gene-set test asks whether the candidate genes have smaller  $p$ -values than genes in general (Table 4).

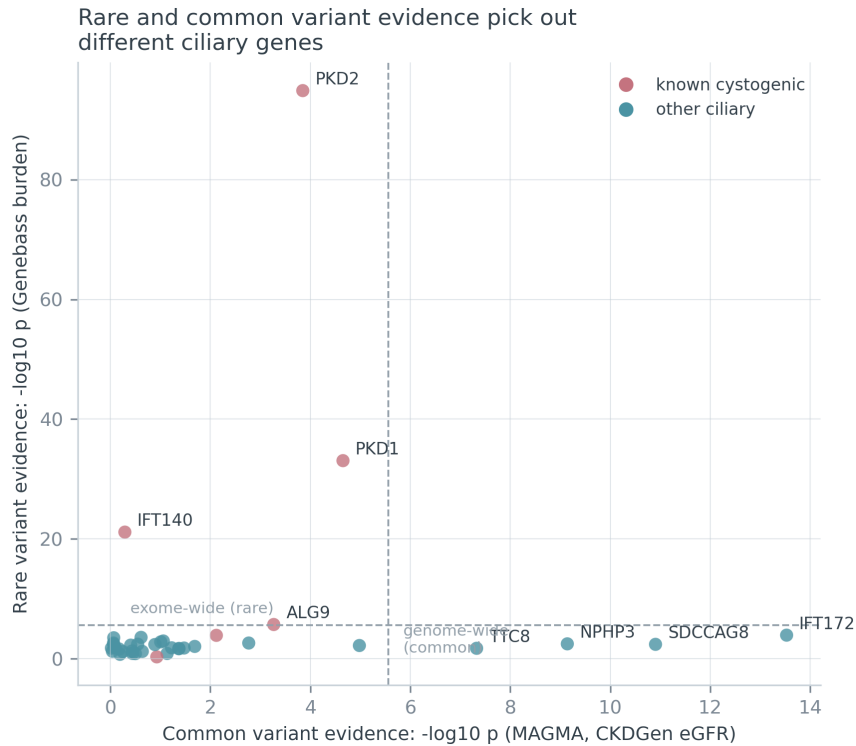
**Table 4:** MAGMA competitive gene-set analysis against estimated glomerular filtration rate. Positive  $\beta$  indicates stronger association than the genome-wide background.

| Gene set                     | Genes | $\beta$ | SE    | $p$   |
|------------------------------|-------|---------|-------|-------|
| Ciliary (all)                | 37    | 0.160   | 0.210 | 0.223 |
| Ciliary (cystogenic removed) | 31    | 0.013   | 0.241 | 0.479 |
| Known cystogenic             | 6     | 0.637   | 0.433 | 0.071 |
| Tier 1                       | 166   | 0.171   | 0.104 | 0.049 |

No gene set is enriched after accounting for the four tests performed. This is a meaningful negative: the ciliary signal identified in the exome data is specific to rare loss-of-function variation and does not extend to common variation.

Two facts must be reported together, because they appear contradictory. Several individual ciliary genes *are* genome-wide significant for eGFR common variants—*IFT172* ( $p = 3.0 \times 10^{-14}$ ), *SDCCAG8* ( $p = 1.2 \times 10^{-11}$ ), *NPHP3* ( $p = 7.2 \times 10^{-10}$ ), *TTC8* ( $p = 4.7 \times 10^{-8}$ ) and *BBS1* ( $p = 1.0 \times 10^{-5}$ ). Yet the set as a whole is not enriched. The reason is that kidney function is extremely polygenic: 1,294 of 18,030 genes (7.2%) pass genome-wide significance in this GWAS. Against that background, five significant ciliary genes is unremarkable. The competitive gene-set test is the correct inference for the question “do ciliary genes matter as a class”; the individual associations remain of interest for follow-up but do not by themselves support the class-level claim.

Figure 14 shows that rare and common variant evidence largely pick out different genes. *PKD1* and *PKD2* dominate the rare-variant axis; *IFT172* and *SDCCAG8* dominate the common-variant axis. This is expected if rare loss-of-function alleles and common regulatory variation act through different genes within the same pathway.



**Figure 14:** Rare-variant burden evidence against common-variant gene-level evidence for the same 37 ciliary genes. The two lines of evidence highlight different genes.

### 3.8 No genetic correlation between CHD and kidney function

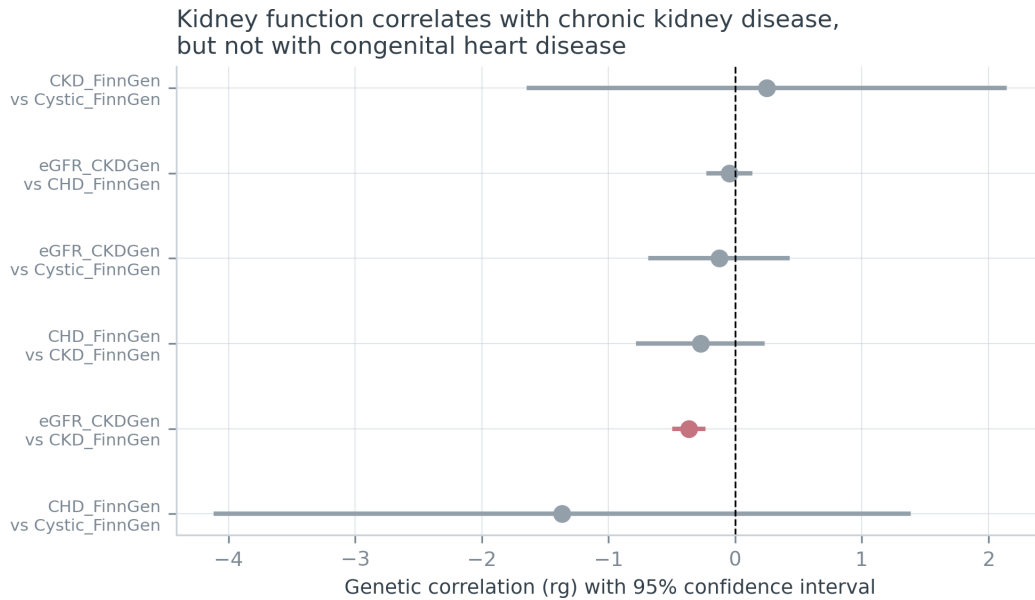
Cross-trait LD score regression gave estimated glomerular filtration rate a SNP heritability of 0.070 (SE 0.0044,  $z = 16.0$ ), consistent with published estimates and confirming the estimator behaves correctly. The positive control performed as expected: eGFR and chronic kidney disease

correlate at  $r_g = -0.365$  (SE 0.057,  $p = 2.0 \times 10^{-10}$ ), the negative sign being correct because higher eGFR means better kidney function.

Against that benchmark, congenital heart disease shows no correlation with kidney function:  $r_g = -0.047$  (SE 0.084,  $p = 0.58$ ; Table 5, Figure 15). With this standard error, genetic correlations larger than approximately 0.23 in absolute value can be excluded at 80% power.

**Table 5:** Cross-trait genetic correlations from LD score regression.

| Trait pair                         | $r_g$  | SE    | $p$                   | 95% CI       |
|------------------------------------|--------|-------|-----------------------|--------------|
| eGFR $\sim$ CKD (positive control) | -0.365 | 0.057 | $2.0 \times 10^{-10}$ | -0.48, -0.25 |
| eGFR $\sim$ CHD                    | -0.047 | 0.084 | 0.58                  | -0.21, 0.12  |
| CHD $\sim$ CKD                     | -0.275 | 0.250 | 0.27                  | -0.77, 0.22  |
| CHD $\sim$ cystic kidney disease   | -1.367 | 1.395 | 0.33                  | -4.10, 1.37  |



**Figure 15:** Genetic correlations with 95% confidence intervals. Only the positive control excludes zero.

### 3.9 Mendelian randomisation finds no causal effect, within limited power

Instrument availability is the binding constraint. At genome-wide significance the FinnGen CHD GWAS yields 67 variants resolving to seven independent regions, but four are Finnish-enriched signals with allele frequencies near 0.001, and one further region (*PALMD*-adjacent 17q21, rs6504672) is absent from CKDGen. After frequency filtering and harmonisation only two instruments remain — rs11166276 near *PALMD* and rs4689912 near *STX18*. Relaxing the threshold to  $p < 10^{-5}$  gives 18 usable instruments.

No analysis shows evidence of a causal effect of congenital heart disease liability on kidney function (Table 6). The two-instrument estimate is  $-0.0004$  (95% CI  $-0.0036$  to  $+0.0028$ ,  $p = 0.81$ ); the eighteen-instrument estimate is  $-0.0009$  (95% CI  $-0.0029$  to  $+0.0012$ ,  $p = 0.41$ ). The weighted median agrees, and the MR-Egger intercept gives no evidence of directional pleiotropy

( $p = 0.16$ ). Leave-one-out analysis moves the estimate by at most 1.17 standard errors, so no single instrument drives the result. All 18 instruments individually favour the assumed causal direction on the Steiger test.

Two caveats limit how much weight this can carry. Cochran’s  $Q$  is significant at the relaxed threshold ( $Q = 42.4$  on 17 df,  $p = 5.9 \times 10^{-4}$ ), indicating that the instruments disagree more than chance allows, as expected when the selection threshold is loosened; random-effects weighting widens the interval accordingly. More importantly, the minimum detectable effect is 0.0029 in log-eGFR units per unit log-odds of CHD liability. This excludes moderate and large effects but not small ones, so the result is an informative bound rather than a demonstration of no effect.

The negative control behaves as required. Using 469 eGFR instruments against congenital heart disease as outcome — a direction that is biologically impossible, since the malformation is complete before any adult kidney measurement exists — returns +0.234 (95% CI  $-0.657$  to  $+1.124$ ,  $p = 0.61$ ).

**Table 6:** Two-sample Mendelian randomisation. Estimates are in log-eGFR units per unit log-odds of CHD liability, except the negative control, which is reversed. MDE is the minimum effect detectable at 80% power.

| Analysis                                             | Method          | $n$ | Estimate | 95% CI           | $p$  |
|------------------------------------------------------|-----------------|-----|----------|------------------|------|
| CHD $\rightarrow$ eGFR<br>( $p < 5 \times 10^{-8}$ ) | IVW             | 2   | −0.0004  | −0.0036, +0.0028 | 0.81 |
|                                                      | Weighted median | 2   | −0.0007  | −0.0051, +0.0036 | 0.74 |
|                                                      | MDE (80%)       | 2   | 0.0046   | —                | —    |
| CHD $\rightarrow$ eGFR<br>( $p < 10^{-5}$ )          | IVW             | 18  | −0.0009  | −0.0029, +0.0012 | 0.41 |
|                                                      | Weighted median | 18  | −0.0000  | −0.0020, +0.0020 | 1.00 |
|                                                      | MR-Egger slope  | 18  | +0.0036  | −0.0030, +0.0103 | 0.28 |
|                                                      | Egger intercept | 18  | −0.00068 | −0.0016, +0.0003 | 0.16 |
|                                                      | MDE (80%)       | 18  | 0.0029   | —                | —    |
| eGFR $\rightarrow$ CHD                               | IVW             | 469 | +0.234   | −0.657, +1.124   | 0.61 |
| (negative control)                                   | Weighted median | 469 | −0.050   | −1.506, +1.406   | 0.95 |

### 3.10 The adjustment survives a flexible model but not a richer one

Doubly robust estimation with gradient-boosting nuisance models separates two distinct threats to the primary analysis (Table 7).

Relaxing the functional form alone leaves both conclusions intact. With the same two covariates as the primary model but no linearity assumption, cardiac involvement raises the probability of a structural kidney anomaly by 3.8 percentage points (95% CI 1.3 to 6.4,  $p = 0.003$ ), while the effect on CKD remains null ( $-0.3$  points,  $p = 0.66$ ). The pattern that motivates this report — malformation yes, chronic disease no — is therefore not an artefact of forcing a straight line through the log-odds.

Adjusting for *which* organ systems are involved is a different matter. The kidney anomaly effect falls to  $-0.011$  and loses significance ( $p = 0.31$ ), and the CKD estimate becomes significantly

negative ( $-0.015$ ,  $p = 0.008$ ). Two things caution against reading this as a refutation. First, the truncated fraction rises from under 2% to roughly 13%, meaning that for one disease in eight the estimator is extrapolating to a comparison the data do not contain — a positivity violation rather than a substantive finding. Second, the other organ-system indicators are themselves consequences of the same shared developmental cause, so conditioning on them is adjustment for descendants of the confounder, which can introduce bias rather than remove it.

The honest reading is therefore that the CAKUT association is robust to how the confounder is modelled but sensitive to how richly it is specified, and that the available annotations cannot settle which specification is correct. The null result for CKD, by contrast, is stable across every specification tried.

**Table 7:** Doubly robust (AIPW) estimates with gradient-boosting nuisance models and five-fold cross-fitting. Estimates are risk differences and are not comparable to the odds ratios above. “Clipped” is the fraction of diseases whose propensity score required truncation, a positivity diagnostic.

| Outcome        | Adjustment                 | Estimate | 95% CI         | Clipped |
|----------------|----------------------------|----------|----------------|---------|
| Kidney anomaly | unadjusted risk difference | +0.195   | —              | —       |
|                | logistic, counts (OR 1.77) | —        | —              | —       |
|                | AIPW, counts only          | +0.038   | +0.013, +0.064 | 1.9%    |
|                | AIPW, counts + each system | −0.011   | −0.031, +0.010 | 13.1%   |
| CKD            | unadjusted risk difference | +0.021   | —              | —       |
|                | logistic, counts (OR 0.95) | —        | —              | —       |
|                | AIPW, counts only          | −0.003   | −0.016, +0.010 | 1.9%    |
|                | AIPW, counts + each system | −0.015   | −0.026, −0.004 | 12.7%   |

## 4. DISCUSSION

### 4.1 The comorbidity is real; the shared genetics are not

Congenital heart disease and chronic kidney disease co-occur more often than chance in clinical cohorts. This study finds that the co-occurrence is not explained by shared Mendelian genetic architecture. Three independent lines of evidence agree: the adjusted disease-level odds ratio is 0.95, the genetic correlation is  $-0.05$ , and no non-cystogenic gene reaches exome-wide significance for chronic renal failure.

The most parsimonious interpretation is that chronic kidney disease in congenital heart disease patients is largely acquired. Chronic cyanosis, altered renal perfusion, contrast nephropathy and cardiopulmonary bypass are each individually plausible [8, 9], and the genetic null result is consistent with all of them.

### 4.2 What does share genetics is CAKUT

The association between congenital heart disease and *structural* kidney malformation survives every adjustment applied (adjusted OR 1.77) and was recovered independently in UK Biobank

exomes, where congenital kidney malformation was the phenotype with the highest proportion of nominally significant ciliary genes. This is developmentally coherent: cilia govern both left–right patterning and nephron induction, so a single ciliary lesion can produce both phenotypes [10, 11].

### 4.3 The subcomplex gradient

That intraflagellar transport and dynein-2 carry signal while the BBSome does not is, to our knowledge, a novel observation in this context and has a natural mechanistic reading. Intraflagellar transport and dynein-2 are required to build and maintain the axoneme; losing them abolishes the organelle. The BBSome by contrast regulates which cargo enters an already-formed cilium. The adult kidney appears to tolerate impaired cargo selectivity but not impaired ciliary assembly. This should be treated as hypothesis-generating: the comparison rests on eight genes per group and none of the individual BBSome genes is significantly null.

### 4.4 Rare and common variation tell different stories

The exome data and the common-variant GWAS agree that kidney function is affected by ciliary genes, but disagree about which ones and at what frequency. Rare loss-of-function burden implicates *PKD1*, *PKD2* and *IFT140*; common-variant gene-level association implicates *IFT172*, *SDCCAG8* and *NPHP3*. Neither picture is wrong. Rare high-impact alleles and common regulatory variation need not converge on the same genes even within a shared pathway, and the competitive gene-set test shows that the ciliary pathway as a whole carries no more common-variant signal than the polygenic background. The class-level claim in this thesis therefore rests on the rare-variant evidence alone.

### 4.5 Survivorship bias as a structural constraint

The lesion-level analysis exposes a limitation that no analytical choice can remedy. Adult biobanks are recruited from survivors, and the cardiac lesions most strongly linked to kidney anomaly are those with the poorest historical survival. Any study of congenital disease in a cohort recruited at ages 40–69 is studying the mild tail of the phenotype distribution. The same reasoning explains why dominant, adult-onset cystogenic genes dominate the exome results while recessive severe ciliopathy genes are silent.

### 4.6 What causal analysis can and cannot add here

Three of the analyses in this report bear on causation, and they fail in informative ways. The pleiotropy adjustment is a backdoor-blocking argument that depends on a proxy standing in for an unmeasured confounder. Mendelian randomisation avoids that dependence entirely but is starved of instruments, because a congenital condition affecting roughly one birth in a hundred yields two usable variants at conventional thresholds. The doubly robust analysis relaxes the

modelling assumptions but runs into positivity: syndromic and non-syndromic diseases are so different that for one disease in eight there is no comparable counterpart.

These are not independent failures. All three trace to the same structural problem noted in Section 4.5: the phenotype that carries the signal is congenital, rare, and largely absent from the adult cohorts where genetic resources are concentrated. No reweighting of the available data resolves that, which is why the causal claims made here are deliberately bounded — an upper limit on effect size, rather than an effect estimate.

## 4.7 Limitations

Several limitations apply. First, the Human Phenotype Ontology is curated from OMIM and Orphanet and therefore describes rare Mendelian disease; it says nothing about common-variant kidney biology, and genes such as *UMOD*, *APOL1* and *SHROOM3* are absent by construction. The LD score regression analysis partly addresses this gap.

Second, adjusting for annotation depth controls one form of ascertainment bias but cannot remove it entirely, since well-studied syndromes are annotated more thoroughly.

Third, the analysis is disease-level rather than patient-level. It measures whether genetic aetiologies co-occur, not how often the two conditions co-occur in living patients.

Fourth, the LD score regression involving congenital heart disease is underpowered. The heritability  $z$ -score for the FinnGen congenital heart disease endpoint is 1.97, below the conventional reliability threshold of 4. The bound excluding  $|r_g| > 0.23$  is valid; a precise point estimate is not.

Fifth, the LD score regression was performed with a reimplementaion of the published estimator rather than the reference software. It was validated against a known heritability estimate and a positive control, but this should be stated in any publication.

Finally, the CKDGen summary statistics are genomic-control corrected, which deflates test statistics; the observed heritability intercept of 0.894 reflects this. Slope-based genetic correlation estimates are unaffected.

## 5. CONCLUSION

Congenital heart disease and chronic kidney disease do not share Mendelian genetic architecture once syndromic pleiotropy is controlled for. The genuine shared axis is between congenital heart disease and congenital anomalies of the kidney and urinary tract, and it is mediated by ciliary biology—specifically by the intraflagellar transport machinery that assembles the cilium, rather than by the BBSome cargo-sorting complex.

For study design this implies three things. Candidate gene lists for this question should be narrow and ciliary rather than broad. The phenotype should be structural kidney anomaly rather than chronic kidney disease. And the outcome should be a quantitative kidney measurement available in the whole cohort rather than a rare binary comorbidity, which in UK Biobank would comprise approximately 18 individuals.

## A. SUPPLEMENTARY INFORMATION

### S1. The 37 ciliary candidate genes

Genes causing both a cardiac and a kidney malformation and assigned to a ciliopathy, nephronophthisis or polycystic kidney disease mechanism class.  $n$  columns give the number of distinct diseases supporting each phenotype. pLI and LOEUF are gnomAD v4.1 loss-of-function constraint metrics; lower LOEUF indicates greater intolerance. The six genes in bold are the established cystogenic genes held out as a positive control.

| Gene           | Ensembl         | Chr | $n$ cardiac | $n$ kidney | $n$ CKD | pLI  | LOEUF |
|----------------|-----------------|-----|-------------|------------|---------|------|-------|
| AHI1           | ENSG00000135541 | 6   | 3           | 1          | 1       | 0.00 | 1.07  |
| <b>ALG9</b>    | ENSG00000086848 | 11  | 4           | 4          | 1       | 0.00 | 0.85  |
| ANKS6          | ENSG00000165138 | 9   | 1           | 1          | 1       | 0.00 | 0.87  |
| BBIP1          | ENSG00000214413 | 10  | 1           | 1          | 2       | 0.07 | 1.69  |
| BBS1           | ENSG00000174483 | 11  | 2           | 1          | 1       | 0.00 | 1.00  |
| BBS10          | ENSG00000179941 | 12  | 1           | 2          | 2       | 0.00 | 1.42  |
| BBS12          | ENSG00000181004 | 4   | 1           | 2          | 1       | 0.00 | 1.42  |
| BBS2           | ENSG00000125124 | 16  | 2           | 3          | 2       | 0.00 | 0.89  |
| BBS4           | ENSG00000140463 | 15  | 1           | 2          | 1       | 0.00 | 0.95  |
| BBS5           | ENSG00000163093 | 2   | 1           | 1          | 1       | 0.00 | 0.88  |
| BBS7           | ENSG00000138686 | 4   | 1           | 1          | 1       | 0.00 | 0.70  |
| BBS9           | ENSG00000122507 | 7   | 1           | 1          | 2       | 0.00 | 0.96  |
| CC2D2A         | ENSG00000048342 | 4   | 1           | 4          | 3       | 0.00 | 0.85  |
| CEP290         | ENSG00000198707 | 12  | 3           | 5          | 6       | 0.00 | 1.09  |
| <b>DNAJB11</b> | ENSG00000090520 | 3   | 1           | 2          | 2       | 0.00 | 0.93  |
| DYNC2I1        | ENSG00000126870 | 7   | 1           | 2          | 1       | 0.00 | 0.80  |
| <b>GANAB</b>   | ENSG00000089597 | 11  | 1           | 2          | 1       | 1.00 | 0.39  |
| IFT122         | ENSG00000163913 | 3   | 1           | 1          | 1       | 0.00 | 0.81  |
| <b>IFT140</b>  | ENSG00000187535 | 16  | 3           | 5          | 5       | 0.00 | 0.97  |
| IFT172         | ENSG00000138002 | 2   | 3           | 3          | 3       | 0.00 | 0.74  |
| INVS           | ENSG00000119509 | 9   | 1           | 2          | 2       | 0.00 | 0.94  |
| LZTFL1         | ENSG00000163818 | 3   | 2           | 2          | 2       | 0.00 | 0.87  |
| MKKS           | ENSG00000125863 | 20  | 2           | 4          | 1       | 0.00 | 1.11  |
| MKS1           | ENSG00000011143 | 17  | 5           | 3          | 1       | 0.00 | 0.99  |
| NEK8           | ENSG00000160602 | 17  | 3           | 5          | 4       | 0.00 | 0.83  |
| NPHP1          | ENSG00000144061 | 2   | 1           | 5          | 6       | 0.00 | 0.92  |
| NPHP3          | ENSG00000113971 | 3   | 2           | 6          | 5       | 0.00 | 0.87  |
| <b>PKD1</b>    | ENSG00000008710 | 16  | 2           | 2          | 2       | 1.00 | 0.36  |
| <b>PKD2</b>    | ENSG00000118762 | 4   | 2           | 2          | 2       | 0.00 | 0.68  |
| RPGRIP1L       | ENSG00000103494 | 16  | 1           | 5          | 4       | 0.00 | 0.91  |
| SDCCAG8        | ENSG00000054282 | 1   | 1           | 4          | 3       | 0.00 | 0.84  |



| Gene    | Ensembl         | Chr | $n$ cardiac | $n$ kidney | $n$ CKD | pLI  | LOEUF |
|---------|-----------------|-----|-------------|------------|---------|------|-------|
| TMEM216 | ENSG00000187049 | 11  | 2           | 4          | 2       | 0.00 | 1.45  |
| TMEM231 | ENSG00000205084 | 16  | 2           | 4          | 1       | 0.00 | 1.41  |
| TMEM237 | ENSG00000155755 | 2   | 3           | 2          | 2       | 0.00 | 1.32  |
| TMEM67  | ENSG00000164953 | 8   | 2           | 9          | 7       | 0.00 | 1.08  |
| TTC8    | ENSG00000165533 | 14  | 2           | 2          | 1       | 0.00 | 0.80  |
| ZNF423  | ENSG00000102935 | 16  | 1           | 1          | 1       | 1.00 | 0.20  |

## S2. SNP heritability estimates

Observed-scale SNP heritability from LD score regression. The  $z$ -score is the heritability divided by its standard error; values below 4 indicate that genetic correlations involving that trait should be treated as underpowered. The intercept estimates confounding from population stratification, and should be close to 1 in a well-controlled study. The value below 1 for eGFR reflects genomic-control correction applied to the published CKDGen summary statistics.

| Trait          | $h^2$  | SE     | $z$   | Intercept | SNPs      |
|----------------|--------|--------|-------|-----------|-----------|
| eGFR_CKDGen    | 0.0702 | 0.0044 | 15.98 | 0.894     | 1,176,043 |
| CHD_FinnGen    | 0.0526 | 0.0267 | 1.97  | 1.019     | 1,158,481 |
| CKD_FinnGen    | 0.0596 | 0.0145 | 4.11  | 1.047     | 1,158,473 |
| Cystic_FinnGen | 0.0169 | 0.1351 | 0.13  | 1.059     | 1,158,476 |

## S3. Renal phenotypes queried in Genebass

Twenty phenotypes were retrieved for each candidate gene: five continuous biomarkers and fifteen ICD-10 based endpoints. Case counts are those analysed by Genebass.

| Type       | Phenotype                                                 | <i>n</i> |
|------------|-----------------------------------------------------------|----------|
| Continuous | Creatinine                                                | 376,624  |
| Continuous | Cystatin C                                                | 376,784  |
| Continuous | Urea                                                      | 376,551  |
| Continuous | Creatinine (enzymatic) in urine                           | 383,774  |
| Continuous | Microalbumin in urine                                     | 119,013  |
| ICD-10     | N18 chronic renal failure                                 | 14,086   |
| ICD-10     | N17 acute renal failure                                   | 7,312    |
| ICD-10     | N20 calculus of kidney and ureter                         | 5,879    |
| ICD-10     | N28 other disorders of kidney and ureter                  | 5,275    |
| ICD-10     | N23 unspecified renal colic                               | 4,304    |
| ICD-10     | N19 unspecified renal failure                             | 2,345    |
| ICD-10     | I12 hypertensive renal disease                            | 1,513    |
| ICD-10     | C64 malignant neoplasm of kidney                          | 985      |
| ICD-10     | Q61 cystic kidney disease                                 | 735      |
| ICD-10     | N08 glomerular disorders in diseases classified elsewhere | 531      |
| ICD-10     | Q63 other congenital malformations of kidney              | 452      |
| ICD-10     | N04 nephrotic syndrome                                    | 242      |
| ICD-10     | Q60 renal agenesis and reduction defects                  | 199      |
| ICD-10     | N26 unspecified contracted kidney                         | 176      |
| ICD-10     | N25 disorders from impaired renal tubular function        | 160      |

#### S4. Sample sizes underlying the power calculations

UK Biobank counts used in Section 3.3 and Figure 9. The congenital heart disease count is the published electronic-health-record ascertainment; the kidney malformation count is the sum of the Q60, Q61 and Q63 first-occurrence fields analysed by Genebass.

| Phenotype                                         | <i>n</i>     |
|---------------------------------------------------|--------------|
| Participants with a creatinine measurement        | 376,624      |
| Chronic kidney disease (N18)                      | 14,086       |
| Congenital heart disease (Q20–Q26)                | 3,385        |
| Congenital kidney malformation (Q60, Q61, Q63)    | 1,386        |
| Expected co-occurrence if independent             | $\approx 10$ |
| Expected co-occurrence at the observed OR of 1.77 | $\approx 18$ |

#### S5. Data and code availability

All data are public and required no application. Analysis code, gene lists, result tables and the source for every figure in this report are available at [github.com/sahoo2000/chd-ckd-genetics](https://github.com/sahoo2000/chd-ckd-genetics). Resource versions are listed in Table 1.

A complete regenie [28] pipeline for individual-level UK Biobank analysis is included in the repository but was *not* executed, as this project had no individual-level access. It is retained because the gene sets and analysis logic are access-independent. Its Python components were verified against synthetic data; its shell components have never been run against real genotype data. This is stated in the repository documentation and should not be presented as executed work.

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